

From Platonic Solids to Regular 4-Polytopes: A Topological and Coxeter-Theoretic Classification with Lean 4 Certificate generated by LLM and registered in Palomar

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Github: <https://github.com/catskillsresearch/platonic-solids>

Abstract

We present an expository and formally verified classification of the regular convex polyhedra in $\mathbb{R}^3$ (the Platonic solids) and the regular convex polychora in $\mathbb{R}^4$ formalized in **Lean 4**. In dimension 3, we formalize the deduction from the topological Euler characteristic of the 2-sphere ($\chi(S^2) = 2$) and incidence combinatorics to the Diophantine inequality $(p-2)(q-2) < 4$, proving the existence of exactly 5 solutions. A key formalization result is the complete elimination of the classical non-degeneracy axiom: the topological and regularity conditions alone algebraically force $E > 0$ without geometric postulation. In dimension 4, where the Euler–Poincaré characteristic vanishes ($\chi(S^3) = 0$) and topology yields no upper bound on flags, we discard all heuristic combinatorial filters and classify the polychora via **Coxeter reflection groups and Gram matrices**. By Sylvester’s criterion, spherical 4-polytopes correspond bijectively to positive-definite Schläfli matrices. We compute the leading principal minors directly and verify that out of the 11 candidate triples formed by Platonic cells and vertex figures, exactly 6 yield positive determinants, 1 yields the parabolic cubic honeycomb ($\det = 0$), and 4 yield hyperbolic honeycombs ($\det < 0$). The accompanying Lean 4 formalization is self-contained, contains **zero axioms, zero admits, and zero sorrys**, deriving all results from first principles. Companion artifact: `PlatonicSolids.lean` (standalone Lake package in this directory).

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1 1. Introduction: The Dimensional Hierarchy

In Euclidean geometry, a regular polytope is a figure whose symmetry group acts transitively on its flags (incident chains of vertices, edges, faces, and higher-dimensional cells).

- In **dimension 2**, there exist infinitely many regular polygons $\{p\}$ for each $p \geq 3$.
- In **dimension 3**, Euclid's *Elements* (Book XIII) establishes that there exist exactly **5 Platonic solids**.
- In **dimension 4**, Ludwig Schläfli proved in 1852 that there exist exactly **6 regular convex 4-polytopes** (polychora).
- In **dimension $n \geq 5$** , the classification collapses permanently to **3 families**: the regular simplex, the hypercube (measure polytope), and the cross-polytope.

This distribution raises foundational mathematical and pedagogical questions: **Why does dimension 3 permit 5 shapes, dimension 4 permit 6 (admitting the exceptional self-dual 24-cell), and dimension $n \geq 5$ permit only 3? Why does topology suffice to bound the classification in 3D, while failing completely in 4D?**

This paper addresses these questions through an analytical exposition coupled with a fully verified formal certificate in Lean 4.

2 2. Dimension 3: Topology, Incidence, and the Non-Degeneracy Insight

2.1 2.1. Topological Derivation of Euler's Formula

The Lean gists are `PlatonicSolids/RadialProjection.lean` (frontier of a convex body $\simeq S^{n-1}$), `PlatonicSolids/EulerPoincare.lean` (algebraic Euler–Poincaré: $\chi(C) = \chi(H_*)$ by rank-nullity), `PlatonicSolids/Sphere2Homology.lean` (cellular $H_*(S^2; \mathbb{Q})$ of the tetrahedron surface), and `PlatonicSolids/EulerBetti.lean` (the numbers $\chi(S^2) = 2$ and $\chi(S^3) = 0$). Let $P \subset \mathbb{R}^3$ be a convex polyhedron containing the origin in its interior. Radial projection $\pi : \partial P \rightarrow S^2$ onto the circumscribed 2-sphere is a homeomorphism. This projection induces a finite CW-complex structure on S^2 with V 0-cells (vertices), E 1-cells (edges), and F 2-cells (faces).

From the cellular homology of the 2-sphere:

$$H_0(S^2; \mathbb{Z}) \cong \mathbb{Z}, \quad H_1(S^2; \mathbb{Z}) \cong 0, \quad H_2(S^2; \mathbb{Z}) \cong \mathbb{Z}$$

The topological Euler characteristic $\chi(S^2)$ is defined as the alternating sum of Betti numbers:

$$\chi(S^2) = b_0 - b_1 + b_2 = 1 - 0 + 1 = 2$$

By the Euler–Poincaré theorem (proved in Lean as rank-nullity on the cellular chain complex: `euler_poincare`), this topological invariant equals the alternating sum of cell counts:

$$V - E + F = 2$$

Radial projection realises ∂P as a topological sphere (`frontier_convex_homeo_sphere`). The Betti numbers 1, 0, 1 are proved for the standard simplicial 2-sphere — the surface of a tetrahedron, with 4 vertices, 6 edges and 4 triangles — by an explicit rank calculation over $\mathbb{Q}$

(`homology_sphere2` in `PlatonicSolids/Sphere2Homology.lean`). Both boundary maps have rank 3, so $b_0 = 4 - 3 = 1$, $b_1 = 3 - 3 = 0$, $b_2 = 4 - 3 = 1$. Euler–Poincaré then gives $V + F = E + 2$ on that complex (`euler_tetrahedron`), and the tetrahedron incidence structure is Platonic (`tetrahedron_platonic`). Mathlib does not compute singular homology of `Metric.sphere`; this is the cellular/simplicial model the paper uses. The general lemma `euler_formula_of_sphere2` still takes Betti numbers as hypotheses so that it applies to any 2-complex with the homology of S^2 .

2.2 2.2. Incidence Combinatorics

The Lean gist is `PlatonicSolids/Incidence.lean` (with the pair predicate in `PlatonicSolids/PlatonicPair.lean`). A regular polyhedron has Schläfli symbol $\{p, q\}$, meaning: 1. Every 2D face is an identical regular p -gon ($p \geq 3$). Each face contributes p edge-face incidences. Since every edge is shared by exactly 2 faces:

$$pF = 2E \implies F = \frac{2E}{p}$$

2. Every vertex meets q faces and q edges ($q \geq 3$). Each vertex contributes q edge-vertex incidences. Since every edge connects exactly 2 vertices:

$$qV = 2E \implies V = \frac{2E}{q}$$

2.3 2.3. The Diophantine Reduction

The Lean gist is `PlatonicSolids/PlatonicPair.lean`; the reciprocal form $1/p + 1/q > 1/2$ is `PlatonicSolids/Reciprocal.lean`. Substituting these incidence relations into Euler’s formula:

$$\frac{2E}{q} - E + \frac{2E}{p} = 2$$

Multiplying by pq over $\mathbb{N}$ avoids rational arithmetic:

$$(2p + 2q)E = (pq)E + 2pq$$

Rearranging terms:

$$(2p + 2q - pq)E = 2pq$$

Because $2pq > 0$ for $p, q \geq 3$, the factor $(2p + 2q - pq)$ must be strictly positive, which yields:

$$pq < 2p + 2q \iff (p - 2)(q - 2) < 4$$

Since $p \geq 3$ and $q \geq 3$, both $(p - 2)$ and $(q - 2)$ are strictly positive integers. The only pairs of positive integers with product strictly less than 4 are:

$(p - 2)$	$(q - 2)$	$(p - 2)(q - 2)$	Pair (p, q)	Polyhedron
1	1	1	(3, 3)	Tetrahedron
1	2	2	(3, 4)	Octahedron
2	1	2	(4, 3)	Cube (Hexahedron)
1	3	3	(3, 5)	Icosahedron
3	1	3	(5, 3)	Dodecahedron

2.4 2.4. Formalization Insight: Elimination of the Non-Degeneracy Axiom

The Lean gist is `edges_pos_of_regular` in `PlatonicSolids/Incidence.lean`. In informal presentations, dividing Euler’s equation by $2E$ to obtain:

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{2} + \frac{1}{E} > \frac{1}{2}$$

is typically justified by appealing to physical or geometric intuition: “*a solid polyhedron clearly has edges, so $E > 0$.*”

When formalizing this theorem in Lean 4, one might initially expect to require an explicit non-degeneracy hypothesis (`hE : E > 0`). However, machine verification reveals this hypothesis to be mathematically redundant.

Recall the cleared identity over $\mathbb{N}$:

$$(2p + 2q)E = (pq)E + 2pq$$

If $E = 0$, the equation reduces to:

$$0 = 2pq$$

Because $p \geq 3$ and $q \geq 3$, we have $2pq \geq 18$, which contradicts $0 = 2pq$.

Therefore, **the topological Euler characteristic ($\chi = 2$) together with the local regularity conditions ($p, q \geq 3$) algebraically prohibits an edgeless configuration.** The condition $E \geq 1$ is an algebraic consequence of the system, not an extrinsic geometric requirement.

3 3. Dimension 4: Why Topology Fails and Coxeter Takes Over

3.1 3.1. The Topological Barrier: $\chi(S^3) = 0$

The Lean gist is `eulerChar_sphere3_betti` in `PlatonicSolids/EulerBetti.lean`. For a convex 4-polytope with boundary homeomorphic to the 3-sphere S^3 , the cellular homology yields:

$$H_0(S^3; \mathbb{Z}) \cong \mathbb{Z}, \quad H_1(S^3; \mathbb{Z}) \cong 0, \quad H_2(S^3; \mathbb{Z}) \cong 0, \quad H_3(S^3; \mathbb{Z}) \cong \mathbb{Z}$$

The Euler–Poincaré characteristic vanishes:

$$\chi(S^3) = b_0 - b_1 + b_2 - b_3 = 1 - 0 + 0 - 1 = 0$$

Hence, the alternating sum of boundary faces satisfies:

$$V - E + F - C = 0$$

Because the alternating sum is zero, dividing by any flag count yields an identity equal to zero rather than a positive reciprocal like $2/E$. **Topology alone does not impose an upper bound on p, q , or r in 4D.**

3.2 3.2. Coxeter Systems and the Schläfli Gram Matrix

The Lean gist is `PlatonicSolids/GramMatrix.lean`. To obtain an upper bound, we use the theory of **Coxeter reflection groups**. A regular 4-polytope is described by its Schläfli symbol $\{p, q, r\}$: * The 3D cells are Platonic solids $\{p, q\}$. * The 3D vertex figures are Platonic solids $\{q, r\}$. * r cells meet around every 1D edge.

The Coxeter group W associated with $\{p, q, r\}$ is generated by reflections $\{s_1, s_2, s_3, s_4\}$ across the facet hyperplanes of a spherical fundamental simplex:

$$W = \langle s_1, s_2, s_3, s_4 \mid (s_i s_j)^{m_{ij}} = 1 \rangle$$

where the exponents are $m_{ii} = 1$, $m_{12} = p$, $m_{23} = q$, $m_{34} = r$, and $m_{ij} = 2$ for $|i - j| > 1$.

The canonical bilinear form of this system is represented by the **Schläfli Gram matrix**:

$$M = \begin{pmatrix} 1 & -\cos \frac{\pi}{p} & 0 & 0 \\ -\cos \frac{\pi}{p} & 1 & -\cos \frac{\pi}{q} & 0 \\ 0 & -\cos \frac{\pi}{q} & 1 & -\cos \frac{\pi}{r} \\ 0 & 0 & -\cos \frac{\pi}{r} & 1 \end{pmatrix}$$

3.3 3.3. Sylvester's Criterion and Principal Minors

The Lean gists are `PlatonicSolids/Angles.lean`, `LeadingMinors.lean`, `TrailingMinor.lean`, `D3IffPlatonic.lean`, `QuadForm.lean`, `CompletingSquare.lean`, and `Sylvester.lean`. By the Cartan–Killing–Coxeter classification: 1. M is **positive definite** ($\det(M) > 0$) $\iff W$ is a **finite reflection group acting on S^3 (a finite regular convex 4-polytope)**. 2. M is **positive semidefinite with nullity 1** ($\det(M) = 0$) $\iff W$ is an **affine reflection group (a Euclidean 3D space-filling honeycomb)**. 3. M has **negative eigenvalues** ($\det(M) < 0$) $\iff W$ is a **hyperbolic reflection group (a honeycomb in hyperbolic space $\mathbb{H}^3$)**.

By Sylvester's criterion, M is positive definite if and only if all leading principal minors $D_k = \det(M_k)$ are strictly positive: * $D_1 = 1 > 0$. * $D_2 = 1 - \cos^2 \frac{\pi}{p} = \sin^2 \frac{\pi}{p} > 0$ (satisfied for all $p \geq 3$). * $D_3 = \sin^2 \frac{\pi}{p} - \cos^2 \frac{\pi}{q}$.

$$D_3 > 0 \iff \sin \frac{\pi}{p} > \cos \frac{\pi}{q} = \sin \left(\frac{\pi}{2} - \frac{\pi}{q} \right) \iff \frac{1}{p} + \frac{1}{q} > \frac{1}{2}$$

Thus, $D_3 > 0$ requires the cell $\{p, q\}$ to be a Platonic solid. * Symmetrically, the trailing 3×3 minor requires:

$$\frac{1}{q} + \frac{1}{r} > \frac{1}{2}$$

Thus, the vertex figure $\{q, r\}$ must also be a Platonic solid.

3.4 3.4. The Critical Dimension-4 Minor: D_4

The Lean gist is `PlatonicSolids/DetFormula.lean`. Expanding $D_4 = \det(M)$ along the fourth row:

$$D_4 = D_3 - \cos^2 \left(\frac{\pi}{r} \right) D_2 = \left(\sin^2 \frac{\pi}{p} - \cos^2 \frac{\pi}{q} \right) - \cos^2 \left(\frac{\pi}{r} \right) \sin^2 \left(\frac{\pi}{p} \right)$$

Factoring $\sin^2 \frac{\pi}{p}$:

$$D_4 = \sin^2 \left(\frac{\pi}{p} \right) \left(1 - \cos^2 \frac{\pi}{r} \right) - \cos^2 \left(\frac{\pi}{q} \right) = \sin^2 \left(\frac{\pi}{p} \right) \sin^2 \left(\frac{\pi}{r} \right) - \cos^2 \left(\frac{\pi}{q} \right)$$

Therefore, the condition that $\{p, q, r\}$ forms a finite regular convex 4-polytope is:

$$\Delta(p, q, r) := \sin^2 \left(\frac{\pi}{p} \right) \sin^2 \left(\frac{\pi}{r} \right) - \cos^2 \left(\frac{\pi}{q} \right) > 0$$

4 4. Evaluation of the 11 Candidate Triples

The Lean gists are `PlatonicSolids/CompatibleTriples.lean` (the eleven triples), `TrigValues.lean` (exact $\sin^2$ and $\cos^2$), `SignTable.lean` (sign of Δ), `Regular4.lean` (the six), and `Classification.lean` (the capstone).

Because both $\{p, q\}$ and $\{q, r\}$ must be Platonic, the indices must satisfy $p, q, r \in \{3, 4, 5\}$. Among the $5 \times 5 = 25$ combinations, only **11 triples** have a compatible middle index q .

The exact squared trigonometric values are:

$$\begin{aligned}\cos^2 \frac{\pi}{3} &= \frac{1}{4}, & \sin^2 \frac{\pi}{3} &= \frac{3}{4} \\ \cos^2 \frac{\pi}{4} &= \frac{1}{2}, & \sin^2 \frac{\pi}{4} &= \frac{1}{2} \\ \cos^2 \frac{\pi}{5} &= \frac{3 + \sqrt{5}}{8}, & \sin^2 \frac{\pi}{5} &= \frac{5 - \sqrt{5}}{8}\end{aligned}$$

Evaluating $\Delta(p, q, r)$ across all 11 candidate triples:

4.1 Group I: Spherical Polychora ($\Delta > 0$)

1. $\{3, 3, 3\}$ (**5-cell** / **Simplex**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{1}{4} = \frac{9}{16} - \frac{4}{16} = \frac{5}{16} > 0$$

2. $\{3, 3, 4\}$ (**16-cell**) & $\{4, 3, 3\}$ (**8-cell** / **Tesseract**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{1}{2}\right) - \frac{1}{4} = \frac{3}{8} - \frac{2}{8} = \frac{1}{8} > 0$$

3. $\{3, 4, 3\}$ (**24-cell** — **Exceptional Self-Dual Polytope**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{1}{2} = \frac{9}{16} - \frac{8}{16} = \frac{1}{16} > 0$$

4. $\{3, 3, 5\}$ (**600-cell**) & $\{5, 3, 3\}$ (**120-cell**):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{5 - \sqrt{5}}{8}\right) - \frac{1}{4} = \frac{7 - 3\sqrt{5}}{32}$$

Since $7^2 = 49 > (3\sqrt{5})^2 = 45$, we have $7 - 3\sqrt{5} > 0$, so $\Delta > 0$.

4.2 Group II: Euclidean Honeycomb ($\Delta = 0$)

5. $\{4, 3, 4\}$ (**Cubic Honeycomb** of $\mathbb{R}^3$):

$$\Delta = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) - \frac{1}{4} = \frac{1}{4} - \frac{1}{4} = 0 \quad (\text{Affine / flat space-filling})$$

4.3 Group III: Hyperbolic Honeycombs ($\Delta < 0$)

6. $\{3, 5, 3\}$ (Icosacell Honeycomb):

$$\Delta = \left(\frac{3}{4}\right) \left(\frac{3}{4}\right) - \frac{3 + \sqrt{5}}{8} = \frac{3 - 2\sqrt{5}}{16} < 0 \quad (\text{since } 3^2 = 9 < 20)$$

7. $\{4, 3, 5\}$ & $\{5, 3, 4\}$:

$$\Delta = \left(\frac{1}{2}\right) \left(\frac{5 - \sqrt{5}}{8}\right) - \frac{1}{4} = \frac{1 - \sqrt{5}}{16} < 0$$

8. $\{5, 3, 5\}$:

$$\Delta = \left(\frac{5 - \sqrt{5}}{8}\right)^2 - \frac{1}{4} = \frac{14 - 10\sqrt{5}}{64} < 0 \quad (\text{since } 14^2 = 196 < 500)$$

Thus, **precisely 6 triples yield positive-definite Gram matrices.**

5 5. Complete Lean 4 Formal Verification

The Lake package compiles against Lean 4 and Mathlib. It contains **zero axioms, zero admits, and zero sorrys** outside the deliberate holes in `Challenge.lean`. The development is split into section-sized modules under `PlatonicSolids/`, each meant to be copied as a gist next to the matching paragraph above. `PlatonicSolids.lean` is the root import. The full sources are reproduced in the appendix.

The compared theorems are:

- `platonic_solids_3d` — Euler $V - E + F = 2$ plus regularity incidence $pF = 2E$, $qV = 2E$ force a Platonic pair (p, q) .
- `edges_pos_of_regular` — the same identities algebraically force $E > 0$; non-degeneracy is a theorem, not an axiom.
- `regular_polychora_classification` — Platonic cells and vertex figures with $\Delta(p, q, r) > 0$ are exactly the six regular convex 4-polytopes.

The stronger equivalence `regular_polychora_iff` identifies those six with positive-definiteness of `schlafliGram` for $p, q, r \geq 3$, via Sylvester’s criterion (`PlatonicSolids/Sylvester.lean`).

`Challenge.lean` restates the compared family with deliberate `sorrys` so Palomar Comparator can check types against `Solution.lean`. Supporting definitions are `IsPlatonicPair`, `IsRegular4Polytope`, `schlafliGram`, and `schlafliDet`.

File	Paper	Content
<code>PlatonicSolids/RadialProjection.lean</code>	§2.1	Frontier of a convex body $\simeq$ sphere
<code>PlatonicSolids/EulerPoincare.lean</code>	§2.1	Algebraic Euler–Poincaré; $V + F = E + 2$ from Betti of S^2

File	Paper	Content
PlatonicSolids/ Sphere2Homology.lean	§2.1	Cellular $H_*(S^2; \mathbb{Q})$ of the tetrahedron surface
PlatonicSolids/ EulerBetti.lean	§§2.1, 3.1	$\chi(S^2) = 2, \chi(S^3) = 0$
PlatonicSolids/ PlatonicPair.lean	§2.3	Five solutions of $(p-2)(q-2) < 4$
PlatonicSolids/ Reciprocal.lean	§§2.3, 3.3	$1/p + 1/q > 1/2$
PlatonicSolids/ Incidence.lean	§§2.2–2.4	platonic_solids_3d, edges_pos_of_regular
PlatonicSolids/ Angles.lean	§3.3	$\pi/n \in (0, \pi/2)$
PlatonicSolids/ GramMatrix.lean	§3.2	schlafliGram, schlafliDet
PlatonicSolids/ LeadingMinors.lean	§3.3	D_1, D_2, D_3
PlatonicSolids/ TrailingMinor.lean	§3.3	Vertex-figure 3×3
PlatonicSolids/ D3IffPlatonic.lean	§3.3	$D_3 > 0$ iff Platonic cell
PlatonicSolids/ QuadForm.lean	§3.3	Gram quadratic form
PlatonicSolids/ CompletingSquare.lean	§3.3	LDL identity
PlatonicSolids/ DetFormula.lean	§3.4	$\det M = \Delta$
PlatonicSolids/ Sylvester.lean	§3.3	PosDef iff leading minors > 0
PlatonicSolids/ TrigValues.lean	§4	$\sin^2(\pi/n), \cos^2(\pi/n)$ for $n \in \{3, 4, 5\}$
PlatonicSolids/ CompatibleTriples.lean	§4	Eleven compatible triples
PlatonicSolids/ SignTable.lean	§4	Sign of Δ
PlatonicSolids/ Regular4.lean	§4	The six regular 4-polytopes
PlatonicSolids/ Classification.lean	§4	Compared 4D theorem and PosDef iff

6 6. Pedagogical and Formalization Significance

1. **Foundational Consistency Across Dimensions:** The mathematical transition between 3D and 4D is made explicit: 3D regular polytopes are bounded by the topology of S^2 , whereas 4D polychora are bounded by the spectral properties of Coxeter reflection groups via Sylvester’s criterion.
2. **Elimination of Redundant Hypotheses:** As demonstrated in `edges_pos_of_regular`, Lean’s linter and tactic engine show that the geometric non-degeneracy condition $E > 0$

is a theorem rather than an independent axiom.

3. **Absence of Ad-Hoc Combinatorial Filters:** Candidate 4D triples are filtered using the **Schläfli determinant** $\Delta(p, q, r) = \sin^2 \frac{\pi}{p} \sin^2 \frac{\pi}{r} - \cos^2 \frac{\pi}{q}$. Every rejection is certified by non-linear arithmetic over real algebraic numbers, with all properties of $\sqrt{5}$ derived from first principles.

7 Acknowledgements

The author expresses sincere gratitude to Professor Dana Scott (Carnegie Mellon University) for suggesting this topic and for inspiring discussions on the formalization of polyhedral geometry. Any remaining errors, misinterpretations, or omissions are solely the author’s own.

7.1 Declaration of Generative AI and AI-Assisted Technologies

In accordance with the Committee on Publication Ethics (COPE) position statements and guidelines on artificial intelligence in research and formal authorship, the author acknowledges the use of large language model tooling during the preparation of this work. Specifically, **Google Gemini 3.8** and **Cursor Grok 4.6** were employed as interactive coding and drafting assistants to assist in exploratory formalization, synthesize initial Lean 4 proof tactics, and help structure the expository drafts. All mathematical claims, Lean 4 proofs, and final text were subsequently reviewed, verified, refined, and certified by the author, who assumes full responsibility for the integrity and accuracy of the content.

8 References

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A Complete Lean source

Checked by lake build against Lean 4 and Mathlib. The development contains no sorry, no admit, and no project-defined axiom. Palomar Comparator compares platonic_solids_3d, edges_pos_of_regular, and regular_polychora_classification.

A.1 PlatonicSolids.lean

Root import of the section-sized modules (33 lines).

```
/-
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Released under Apache 2.0 license as described in the file LICENSE.
Authors: Lars Warren Ericson.
Github: https://github.com/catskillsresearch/platonic-solids/blob/main/PlatonicSolids.lean
-/
import PlatonicSolids.Angles
import PlatonicSolids.Classification
import PlatonicSolids.CompatibleTriples
import PlatonicSolids.CompletingSquare
import PlatonicSolids.D3IffPlatonic
import PlatonicSolids.DetFormula
import PlatonicSolids.EulerBetti
import PlatonicSolids.EulerPoincare
import PlatonicSolids.GramMatrix
import PlatonicSolids.Incidence
import PlatonicSolids.LeadingMinors
import PlatonicSolids.PlatonicPair
import PlatonicSolids.QuadForm
import PlatonicSolids.RadialProjection
import PlatonicSolids.Reciprocal
import PlatonicSolids.Regular4
import PlatonicSolids.SignTable
import PlatonicSolids.Sphere2Homology
import PlatonicSolids.Sylvester
import PlatonicSolids.TrailingMinor
import PlatonicSolids.TrigValues

/-!
Root import of the section-sized modules. Each file under `PlatonicSolids/`
is meant to be readable next to the corresponding paragraph of
`PlatonicSolids.md`.
-/
```

A.2 PlatonicSolids/RadialProjection.lean

Frontier of a bounded convex body is homeomorphic to the sphere (25 lines).

```
/-
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-/
import Mathlib.Analysis.Convex.GaugeRescale

/-!
Paper §2.1. Radial projection: the frontier of a bounded convex body
with nonempty interior is homeomorphic to the unit sphere. Mathlib's
`gaugeRescaleHomeomorph` is the homeomorphism of the ambient space
that carries the frontier onto `Metric.sphere 0 1`.
-/

open Metric Set
```

```

/-- The frontier of a bounded convex set with nonempty interior is a
topological sphere. This is the radial projection of the paper, stated
for a body that need not be centred at the origin. -/
theorem frontier_convex_homeo_sphere
  {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] {s : Set E}
  (hc : Convex ℝ s) (hne : (interior s).Nonempty) (hb : Bornology.IsBounded s) :
  ∃ h : E ≃ₜ E, h '' frontier s = sphere (0 : E) 1 :=
  let ⟨h, _, _, hf⟩ :=
    exists_homeomorph_image_interior_closure_frontier_eq_unitBall hc hne hb
  ⟨h, hf⟩

```

A.3 PlatonicSolids/EulerPoincare.lean

Algebraic Euler–Poincaré: $\chi(C) = \chi(H_*)$ by rank-nullity (68 lines).

```

/-
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-/
import Mathlib.LinearAlgebra.FiniteDimensional.Lemmas

/-!
Paper §2.1. Algebraic Euler–Poincaré for a 2-dimensional chain complex
of finite-dimensional vector spaces (the cellular complex of a polyhedron):

`χ(C) = rank C₀ - rank C₁ + rank C₂` equals `χ(H) = b₀ - b₁ + b₂`.

The identity is rank-nullity. Specializing to the Betti numbers of `S²`
(`b₀ = 1`, `b₁ = 0`, `b₂ = 1`, computed for the tetrahedron surface
in `PlatonicSolids/Sphere2Homology.lean`) yields `V - E + F = 2`,
hence `V + F = E + 2`.
-/

open Module

variable {K : Type*} [Field K]
variable {C₂ C₁ C₀ : Type*}
  [AddCommGroup C₂] [Module K C₂] [FiniteDimensional K C₂]
  [AddCommGroup C₁] [Module K C₁] [FiniteDimensional K C₁]
  [AddCommGroup C₀] [Module K C₀]

/-- `b₀ = rank(C₀ / im d₁)` -/
noncomputable def betti0 (d₁ : C₁ →₁[K] C₀) : ℤ :=
  finrank K C₀ - finrank K (LinearMap.range d₁)

/-- `b₁ = rank(ker d₁ / im d₂)` -/
noncomputable def betti1 (d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀) : ℤ :=
  finrank K (LinearMap.ker d₁) - finrank K (LinearMap.range d₂)

/-- `b₂ = rank(ker d₂)` -/
noncomputable def betti2 (d₂ : C₂ →₁[K] C₁) : ℤ :=
  finrank K (LinearMap.ker d₂)

/-- Algebraic Euler–Poincaré: cell Euler characteristic equals homological
Euler characteristic for any 2-complex with `d₁.comp d₂ = 0`. -/
theorem euler_poincare (d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀)
  (hcomp : d₁.comp d₂ = 0) :
  (finrank K C₀ : ℤ) - finrank K C₁ + finrank K C₂ =
    betti0 d₁ - betti1 d₂ d₁ + betti2 d₂ := by
  have _hB : LinearMap.range d₂ ≤ LinearMap.ker d₁ :=
    LinearMap.range_le_ker_iff.mpr hcomp
  have hr1 : (finrank K (LinearMap.range d₁) : ℤ) + finrank K (LinearMap.ker d₁) =

```

```

    finrank K C1 := by exact_mod_cast d1.finrank_range_add_finrank_ker
  have hr2 : (finrank K (LinearMap.range d2) : ℤ) + finrank K (LinearMap.ker d2) =
    finrank K C2 := by exact_mod_cast d2.finrank_range_add_finrank_ker
  simp only [betti0, betti1, betti2]
  linarith

/-- `V - E + F = 2` is the same counting identity as `V + F = E + 2`. -/
theorem euler_count {V E F : ℕ} (h : (V : ℤ) - E + F = 2) : V + F = E + 2 := by
  omega

/-- Cellular ranks `V, E, F` and the Betti numbers of `S2` give Euler's formula. -/
theorem euler_formula_of_sphere2
  (d2 : C2 →1[K] C1) (d1 : C1 →1[K] C0)
  {V E F : ℕ}
  (hV : finrank K C0 = V) (hE : finrank K C1 = E) (hF : finrank K C2 = F)
  (hcomp : d1.comp d2 = 0)
  (hb0 : betti0 d1 = 1) (hb1 : betti1 d2 d1 = 0) (hb2 : betti2 d2 = 1) :
  V + F = E + 2 := by
  have hχ := euler_poincare d2 d1 hcomp
  rw [hV, hE, hF, hb0, hb1, hb2] at hχ
  exact euler_count hχ

```

A.4 PlatonicSolids/Sphere2Homology.lean

Cellular $H_*(S^2; \mathbb{Q})$ of the tetrahedron surface (129 lines).

```

/-
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-/
import Mathlib.LinearAlgebra.Matrix.Notation
import Mathlib.LinearAlgebra.Matrix.Rank
import Mathlib.LinearAlgebra.Matrix.ToLin
import Mathlib.Tactic
import PlatonicSolids.EulerPoincare
import PlatonicSolids.Incidence

/-!
Paper §2.1. Cellular homology of the standard simplicial  $(S^2)$ :
the boundary of a tetrahedron (4 vertices, 6 edges, 4 triangles).

This is the usual CW / simplicial calculation
`H0 ≅ ℚ`, `H1 = 0`, `H2 ≅ ℚ`. Mathlib's singular homology of
`Metric.sphere` is not computed; this is the chain complex the
Euler-Poincaré argument uses.
-/

open scoped Matrix
open Matrix Module

/-- Oriented incidence vertices ← edges: columns `01,02,03,12,13,23`. -/
def tetra_d1 : Matrix (Fin 4) (Fin 6) ℚ :=
  !![ -1, -1, -1,  0,  0,  0;
      1,  0,  0, -1, -1,  0;
      0,  1,  0,  1,  0, -1;
      0,  0,  1,  0,  1,  1]

/-- Oriented incidence edges ← faces: columns `012,013,023,123`. -/
def tetra_d2 : Matrix (Fin 6) (Fin 4) ℚ :=
  !![ 1,  1,  0,  0;
     -1,  0,  1,  0;
      0, -1, -1,  0;
      1,  0,  0,  1;

```

```

0, 1, 0, -1;
0, 0, 1, 1]

lemma tetra_comp : tetra_d1 * tetra_d2 = 0 := by
  ext i j
  fin_cases i <;> fin_cases j <;> simp [tetra_d1, tetra_d2]

noncomputable def tetra_del1 : (Fin 6 → ℚ) →1[ℚ] (Fin 4 → ℚ) := tetra_d1.toLin'
noncomputable def tetra_del2 : (Fin 4 → ℚ) →1[ℚ] (Fin 6 → ℚ) := tetra_d2.toLin'

lemma tetra_del_comp : tetra_del1.comp tetra_del2 = 0 := by
  rw [tetra_del1, tetra_del2, ← toLin'_mul, tetra_comp]
  exact map_zero _

lemma tetra_d1_rank_ge : 3 ≤ tetra_d1.rank := by
  have hI : ((1 : Matrix (Fin 3) (Fin 3) ℚ).rank) = 3 := rank_one
  have hsub :
    tetra_d1.submatrix ![1, 2, 3] ![0, 1, 2] = (1 : Matrix (Fin 3) (Fin 3) ℚ) := by
    ext i j
    fin_cases i <;> fin_cases j <;> simp [tetra_d1]
  have : (tetra_d1.submatrix ![1, 2, 3] ![0, 1, 2]).rank = 3 := by
    rw [hsub]; exact hI
  exact this.symm.trans_le (rank_submatrix_le _ _ _)

lemma tetra_d2_rank_ge : 3 ≤ tetra_d2.rank := by
  have hI : ((1 : Matrix (Fin 3) (Fin 3) ℚ).rank) = 3 := rank_one
  have hsub :
    tetra_d2.submatrix ![3, 4, 5] ![0, 1, 2] = (1 : Matrix (Fin 3) (Fin 3) ℚ) := by
    ext i j
    fin_cases i <;> fin_cases j <;> simp [tetra_d2]
  have : (tetra_d2.submatrix ![3, 4, 5] ![0, 1, 2]).rank = 3 := by
    rw [hsub]; exact hI
  exact this.symm.trans_le (rank_submatrix_le _ _ _)

lemma tetra_d1_rank : tetra_d1.rank = 3 := by
  have hsum := rank_add_rank_le_card_of_mul_eq_zero tetra_comp
  have h1 := tetra_d1_rank_ge
  have h2 := tetra_d2_rank_ge
  simp only [Fintype.card_fin] at hsum
  omega

lemma tetra_d2_rank : tetra_d2.rank = 3 := by
  have hsum := rank_add_rank_le_card_of_mul_eq_zero tetra_comp
  have h1 := tetra_d1_rank_ge
  have h2 := tetra_d2_rank_ge
  simp only [Fintype.card_fin] at hsum
  omega

lemma tetra_del1_range : finrank ℚ (LinearMap.range tetra_del1) = 3 := by
  simp [tetra_del1, toLin'_apply] using tetra_d1_rank

lemma tetra_del2_range : finrank ℚ (LinearMap.range tetra_del2) = 3 := by
  simp [tetra_del2, toLin'_apply] using tetra_d2_rank

/--  $H_0(S^2; \mathbb{Q}) \cong \mathbb{Q}$ . -/
theorem sphere2_betti0 : betti0 tetra_del1 = 1 := by
  simp [betti0, tetra_del1_range, finrank_fintype_fun_eq_card]

/--  $H_1(S^2; \mathbb{Q}) = 0$ . -/
theorem sphere2_betti1 : betti1 tetra_del2 tetra_del1 = 0 := by
  have hr := tetra_del1.finrank_range_add_finrank_ker
  simp [betti1, tetra_del2_range, tetra_del1_range, finrank_fintype_fun_eq_card] at hr
  omega

```

```

/--  $H_2(S^2; \mathbb{Q}) \cong \mathbb{Q}$ . -/
theorem sphere2_betti2 : betti2 tetra_del2 = 1 := by
  have hr := tetra_del2.finrank_range_add_finrank_ker
  simp [betti2, tetra_del2_range, finrank_fintype_fun_eq_card] at hr ⊢
  omega

/-- Cellular homology of the standard simplicial 2-sphere. -/
theorem homology_sphere2 :
  betti0 tetra_del1 = 1 ∧ betti1 tetra_del2 tetra_del1 = 0 ∧ betti2 tetra_del2 = 1 :=
  ⟨sphere2_betti0, sphere2_betti1, sphere2_betti2⟩

/-- Euler's formula on the tetrahedron, from its computed Betti numbers. -/
theorem euler_tetrahedron : (4 : ℕ) + 4 = 6 + 2 :=
  euler_formula_of_sphere2 tetra_del2 tetra_del1
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  tetra_del_comp sphere2_betti0 sphere2_betti1 sphere2_betti2

/-- The tetrahedron incidence structure is Platonic, using homology not a
raw Euler hypothesis. -/
theorem tetrahedron_platonic : IsPlatonicPair 3 3 :=
  platonic_solids_3d_of_homology tetra_del2 tetra_del1 4 6 4 3 3
  (by decide) (by decide) (by decide) (by decide)
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  (by simp [finrank_fintype_fun_eq_card])
  tetra_del_comp sphere2_betti0 sphere2_betti1 sphere2_betti2

```

A.5 PlatonicSolids/EulerBetti.lean

Betti arithmetic of $\chi(S^2) = 2$ and $\chi(S^3) = 0$ (24 lines).

```

/-
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-/
import Mathlib.Tactic

/-!
Paper §2.1 / §3.1. Betti arithmetic of  $\chi(S^2)$  and  $\chi(S^3)$ .

The algebraic Euler-Poincaré identity (cell ranks equal Betti ranks)
is `euler_poincare` in `PlatonicSolids/EulerPoincare.lean`. The
numbers `b₀ = 1`, `b₁ = 0`, `b₂ = 1` for the standard simplicial
`S²` (tetrahedron surface) are computed in
`PlatonicSolids/Sphere2Homology.lean`. These lemmas record the
resulting  $\chi(S^2) = 2$  and the 4D vanishing  $\chi(S^3) = 0$  (Mathlib
does not compute  $H_*(S^3)$ ), which is why topology supplies no 4D
bound.
-/

/--  $\chi(S^2) = b_0 - b_1 + b_2 = 1 - 0 + 1$ . -/
lemma eulerChar_sphere2_betti : (1 : ℤ) - 0 + 1 = 2 := by norm_num

/--  $\chi(S^3) = b_0 - b_1 + b_2 - b_3 = 1 - 0 + 0 - 1$ . -/
lemma eulerChar_sphere3_betti : (1 : ℤ) - 0 + 0 - 1 = 0 := by norm_num

```

A.6 PlatonicSolids/PlatonicPair.lean

The five solutions of the Platonic Diophantine inequality (38 lines).

```

/-
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-/
import Mathlib.Tactic

/-!
Paper §2.3. The five solutions of  $(p-2)(q-2) < 4$  for  $p, q \geq 3$ .
-/

/-- Schläfli pairs of the five Platonic solids. -/
def IsPlatonicPair (p q : ℕ) : Prop :=
  (p = 3 ∧ q = 3) ∨
  (p = 3 ∧ q = 4) ∨
  (p = 4 ∧ q = 3) ∨
  (p = 3 ∧ q = 5) ∨
  (p = 5 ∧ q = 3)

lemma pq_ge_two_p_add_two_q {p q : ℕ} (hp : 4 ≤ p) (hq : 4 ≤ q) :
  2 * p + 2 * q ≤ p * q := by
  zify
  nlinarith

/-- The inequality  $pq < 2p + 2q$  for  $p, q \geq 3$  has exactly the five Platonic solutions. -/
theorem platonic_pairs_of_inequality {p q : ℕ}
  (hp : 3 ≤ p) (hq : 3 ≤ q) (h : p * q < 2 * p + 2 * q) :
  IsPlatonicPair p q := by
  rcases show p = 3 ∨ q = 3 ∨ (4 ≤ p ∧ 4 ≤ q) by omega with rfl | rfl | ⟨hp4, hq4⟩
  · rcases show q = 3 ∨ q = 4 ∨ q = 5 by omega with rfl | rfl | rfl <;> simp [IsPlatonicPair]
  · rcases show p = 3 ∨ p = 4 ∨ p = 5 by omega with rfl | rfl | rfl <;> simp [IsPlatonicPair]
  · exact (lt_irrefl _ (h.trans_le (pq_ge_two_p_add_two_q hp4 hq4))).elim

theorem IsPlatonicPair.ge_three {p q : ℕ} (h : IsPlatonicPair p q) : 3 ≤ p ∧ 3 ≤ q := by
  rcases h with ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ <;> decide

theorem IsPlatonicPair.inequality {p q : ℕ} (h : IsPlatonicPair p q) :
  p * q < 2 * p + 2 * q := by
  rcases h with ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ <;> decide

```

A.7 PlatonicSolids/Reciprocal.lean

$1/p + 1/q > 1/2$ iff the integer inequality (44 lines).

```

/-
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-/
import Mathlib.Data.Real.Basic
import PlatonicSolids.PlatonicPair

/-!
Paper §2.3 / §3.3.  $1/p + 1/q > 1/2$  is the reciprocal form of the
Platonic Diophantine inequality, and is the condition  $D_3 > 0$ .
-/

lemma reciprocal_iff {p q : ℕ} (hp : 0 < p) (hq : 0 < q) :
  (1 / (p : ℝ) + 1 / q > 1 / 2) ↔ p * q < 2 * p + 2 * q := by
  have : (0 : ℝ) < p := Nat.cast_pos.mpr hp
  have : (0 : ℝ) < q := Nat.cast_pos.mpr hq
  constructor
  · intro h
    have h' : (p + q : ℝ) / (p * q) > 1 / 2 := by
      convert h using 1

```

```

    field_simp
    ring
    rw [gt_iff_lt, lt_div_iff0 (by positivity)] at h'
    exact_mod_cast (by linarith : (p * q : ℝ) < 2 * p + 2 * q)
  · intro h
    have hR : (p * q : ℝ) < 2 * p + 2 * q := mod_cast h
    have h' : (p + q : ℝ) / (p * q) > 1 / 2 := by
      rw [gt_iff_lt, lt_div_iff0 (by positivity)]
      linarith
    convert h' using 1
    field_simp
    ring

theorem platonic_pair_iff {p q : ℕ} :
  IsPlatonicPair p q ↔ 3 ≤ p ∧ 3 ≤ q ∧ p * q < 2 * p + 2 * q := by
  constructor
  · intro h
    exact ⟨h.ge_three.1, h.ge_three.2, h.inequality⟩
  · exact fun ⟨hp, hq, h⟩ => platonic_pairs_of_inequality hp hq h

theorem platonic_pair_iff_reciprocal {p q : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) :
  IsPlatonicPair p q ↔ 1 / (p : ℝ) + 1 / q > 1 / 2 := by
  rw [platonic_pair_iff, reciprocal_iff (by omega) (by omega)]
  simp [hp, hq]

```

A.8 PlatonicSolids/Incidence.lean

Regularity incidence plus Euler force a Platonic pair and $E > 0$ (81 lines).

```

/-
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-/

import PlatonicSolids.EulerPoincare
import PlatonicSolids.PlatonicPair

/-!
Paper §2.2-2.4. Regularity incidence `pF = 2E`, `qV = 2E` and Euler
`V + F = E + 2` force a Platonic pair, and force `E > 0` with no extra
non-degeneracy axiom.
-/

/-- The Euler + regularity identities algebraically prohibit `E = 0`. -/
theorem edges_pos_of_regular
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  0 < E := by
  have h_lhs : p * q * (V + F) = (2 * p + 2 * q) * E := calc
    p * q * (V + F) = p * (q * V) + q * (p * F) := by ring
    _ = p * (2 * E) + q * (2 * E) := by rw [h_edges_verts, h_edges_faces]
    _ = (2 * p + 2 * q) * E := by ring
  have h_rhs : p * q * (V + F) = p * q * E + 2 * p * q := calc
    p * q * (V + F) = p * q * (E + 2) := by rw [h_euler]
    _ = p * q * E + 2 * p * q := by ring
  have h_eq : (2 * p + 2 * q) * E = p * q * E + 2 * p * q := by rw [← h_lhs, h_rhs]
  by_contra hz
  have hE0 : E = 0 := Nat.eq_zero_of_not_pos hz
  subst hE0
  simp only [mul_zero, zero_add] at h_eq
  exact (Nat.pos_iff_ne_zero.mp (by positivity : 0 < 2 * p * q)) h_eq.symm

```

```

/-- Any regular incidence structure with  $\chi = 2$  is a Platonic pair. -/
theorem platonic_solids_3d
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  IsPlatonicPair p q := by
  have h_lhs : p * q * (V + F) = (2 * p + 2 * q) * E := calc
    p * q * (V + F) = p * (q * V) + q * (p * F) := by ring
    _ = p * (2 * E) + q * (2 * E) := by rw [h_edges_verts, h_edges_faces]
    _ = (2 * p + 2 * q) * E := by ring
  have h_eq : (2 * p + 2 * q) * E = p * q * E + 2 * p * q := by
    calc (2 * p + 2 * q) * E = p * q * (V + F) := h_lhs.symm
    _ = p * q * (E + 2) := by rw [h_euler]
    _ = p * q * E + 2 * p * q := by ring
  have h_ineq : p * q < 2 * p + 2 * q := by
    obtain hlt | hge := lt_or_ge (p * q) (2 * p + 2 * q)
    · exact hlt
    · have : (2 * p + 2 * q) * E ≤ p * q * E := Nat.mul_le_mul_right E hge
      have : 0 < 2 * p * q := by positivity
      omega
  exact platonic_pairs_of_inequality hp hq h_ineq

/-- The 3D classification from a cellular 2-complex with the Betti
numbers of  $S^2$ . Euler-Poincaré (euler_formula_of_sphere2) supplies
 $V + F = E + 2$ ; regularity incidence then forces a Platonic pair. -/
theorem platonic_solids_3d_of_homology
  {K : Type*} [Field K]
  {C₂ C₁ C₀ : Type*}
  [AddCommGroup C₂] [Module K C₂] [FiniteDimensional K C₂]
  [AddCommGroup C₁] [Module K C₁] [FiniteDimensional K C₁]
  [AddCommGroup C₀] [Module K C₀] [FiniteDimensional K C₀]
  (d₂ : C₂ →₁[K] C₁) (d₁ : C₁ →₁[K] C₀)
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (hV : Module.finrank K C₀ = V)
  (hE : Module.finrank K C₁ = E)
  (hF : Module.finrank K C₂ = F)
  (hcomp : d₁.comp d₂ = 0)
  (hb0 : betti0 d₁ = 1) (hb1 : betti1 d₂ d₁ = 0) (hb2 : betti2 d₂ = 1) :
  IsPlatonicPair p q :=
  platonic_solids_3d V E F p q hp hq h_edges_faces h_edges_verts
  (euler_formula_of_sphere2 d₂ d₁ hV hE hF hcomp hb0 hb1 hb2)

```

A.9 PlatonicSolids/Angles.lean

$\pi/n \in (0, \pi/2)$ so sine and cosine are positive (28 lines).

/-

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-/

import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic

/-!

Paper §3.3. For $n \geq 3$, the Schläfli angle π/n lies in $(0, \pi/2)$, so $\sin$ and $\cos$ are positive.

-/

```

open Real

lemma angle_pos {n : ℕ} (hn : 1 ≤ n) : 0 < π / n := by
  have : (0 : ℝ) < n := Nat.cast_pos.mpr hn
  positivity

lemma angle_lt_pi_div_two {n : ℕ} (hn : 3 ≤ n) : π / n < π / 2 :=
  div_lt_div_of_pos_left pi_pos (by positivity : (0 : ℝ) < 2)
  (by exact_mod_cast (lt_of_lt_of_le (by decide : (2 : ℕ) < 3) hn))

lemma sin_angle_pos {n : ℕ} (hn : 3 ≤ n) : 0 < sin (π / n) :=
  sin_pos_of_pos_of_lt_pi (angle_pos (Nat.succ_le_of_lt (Nat.zero_lt_of_lt hn)))
  (angle_lt_pi_div_two hn |>.trans (by linarith [pi_pos]))

lemma cos_angle_pos {n : ℕ} (hn : 3 ≤ n) : 0 < cos (π / n) :=
  cos_pos_of_mem_Ioo (by linarith [angle_pos (Nat.succ_le_of_lt (Nat.zero_lt_of_lt hn)), pi_pos
  ],
  angle_lt_pi_div_two hn)

```

A.10 PlatonicSolids/GramMatrix.lean

Schläfli Gram matrix and the scalar Δ (30 lines).

```

/-
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-/
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic
import Mathlib.LinearAlgebra.Matrix.Hermitian
import Mathlib.LinearAlgebra.Matrix.Notation

/-!
Paper §3.2. The Schläfli Gram matrix of  $\{p, q, r\}$  and the scalar
 $\Delta(p, q, r)$  that will equal its determinant.
-/

open scoped Matrix
open Real Matrix

/-- Canonical bilinear form of the rank-4 Schläfli Coxeter system  $\{p, q, r\}$ . -/
noncomputable def schlaflGram (p q r : ℕ) : Matrix (Fin 4) (Fin 4) ℝ :=
  !![1, -cos (π / p), 0, 0;
    -cos (π / p), 1, -cos (π / q), 0;
    0, -cos (π / q), 1, -cos (π / r);
    0, 0, -cos (π / r), 1]

/--  $\Delta(p, q, r) = \sin^2(\pi/p) \sin^2(\pi/r) - \cos^2(\pi/q)$ . -/
noncomputable def schlafliDet (p q r : ℕ) : ℝ :=
  sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2

lemma schlaflGram_isHermitian (p q r : ℕ) : (schlaflGram p q r).IsHermitian := by
  ext i j
  fin_cases i <;> fin_cases j <;> simp [schlaflGram, conjTranspose]

```

A.11 PlatonicSolids/LeadingMinors.lean

Closed forms of the leading minors D_1, D_2, D_3 (34 lines).

```

/-
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-/

```

```

import Mathlib.LinearAlgebra.Matrix.Determinant.Basic
import PlatonicSolids.GramMatrix

/-!
Paper §3.3-3.4. Leading principal minors  $D_1$ ,  $D_2$ ,  $D_3$  of the Gram
matrix, matching the paper's closed forms.
-/

open scoped Matrix
open Real Matrix

def leadingSubmatrix (k : ℕ) (hk : k ≤ 4) (A : Matrix (Fin 4) (Fin 4) ℝ) :
  Matrix (Fin k) (Fin k) ℝ :=
  A.submatrix (Fin.castLE hk) (Fin.castLE hk)

noncomputable def leadingMinor (k : ℕ) (hk : k ≤ 4) (p q r : ℕ) : ℝ :=
  (leadingSubmatrix k hk (schlafligram p q r)).det

lemma leadingMinor_one (p q r : ℕ) : leadingMinor 1 (by decide) p q r = 1 := by
  simp [leadingMinor, leadingSubmatrix, schlafligram]

lemma leadingMinor_two (p q r : ℕ) :
  leadingMinor 2 (by decide) p q r = sin (π / p) ^ 2 := by
  simp [leadingMinor, leadingSubmatrix, schlafligram, det_fin_two]
  linarith [sin_sq_add_cos_sq (π / p)]

lemma leadingMinor_three (p q r : ℕ) :
  leadingMinor 3 (by decide) p q r = sin (π / p) ^ 2 - cos (π / q) ^ 2 := by
  simp [leadingMinor, leadingSubmatrix, schlafligram, det_fin_three]
  linarith [sin_sq_add_cos_sq (π / p)]

```

A.12 PlatonicSolids/TrailingMinor.lean

Trailing 3×3 minor is the vertex-figure cell minor (22 lines).

```

/-
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-/

import Mathlib.LinearAlgebra.Matrix.Determinant.Basic
import PlatonicSolids.GramMatrix

/-!
Paper §3.3. The trailing  $3 \times 3$  principal minor is the cell minor of the
vertex figure  $\{q, r\}$ .
-/

open scoped Matrix
open Real Matrix

noncomputable def trailing3 (p q r : ℕ) : Matrix (Fin 3) (Fin 3) ℝ :=
  (schlafligram p q r).submatrix (fun i : Fin 3 => i.succ) (fun i => i.succ)

lemma trailing3_det (p q r : ℕ) :
  (trailing3 p q r).det = sin (π / q) ^ 2 - cos (π / r) ^ 2 := by
  simp [trailing3, schlafligram, det_fin_three]
  linarith [sin_sq_add_cos_sq (π / q)]

```

A.13 PlatonicSolids/D3IffPlatonic.lean

$D_3 > 0$ iff the cell is Platonic (58 lines).

```

/-
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-/
import PlatonicSolids.Angles
import PlatonicSolids.LeadingMinors
import PlatonicSolids.Reciprocal
import PlatonicSolids.TrailingMinor

/-!
Paper §3.3. `D₃ > 0` iff the cell is Platonic; the trailing 3×3 minor
is the same statement for the vertex figure.
-/

open Real

lemma angle_mem_Icc {n : ℕ} (hn : 3 ≤ n) :
  π / n ∈ Set.Icc (-(π / 2)) (π / 2) :=
  ⟨by linarith [angle_pos (Nat.succ_le_of_lt (Nat.zero_lt_of_lt hn)), pi_pos],
    (angle_lt_pi_div_two hn).le⟩

lemma complement_mem_Icc {n : ℕ} (hn : 3 ≤ n) :
  π / 2 - π / n ∈ Set.Icc (-(π / 2)) (π / 2) :=
  ⟨by linarith [angle_lt_pi_div_two hn, pi_pos],
    by linarith [angle_pos (Nat.succ_le_of_lt (Nat.zero_lt_of_lt hn))]⟩

/-- `sin(π/p) > cos(π/q)` iff `1/p + 1/q > 1/2`. -/
lemma sin_gt_cos_iff_reciprocal {p q : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) :
  sin (π / p) > cos (π / q) ↔ 1 / (p : ℝ) + 1 / q > 1 / 2 := by
  have hp0 : (0 : ℝ) < p := Nat.cast_pos.mpr (Nat.zero_lt_of_lt hp)
  have hq0 : (0 : ℝ) < q := Nat.cast_pos.mpr (Nat.zero_lt_of_lt hq)
  rw [← sin_pi_div_two_sub (π / q), gt_iff_lt, gt_iff_lt]
  rw [strictMonoOn_sin.lt_iff_lt (complement_mem_Icc hq) (angle_mem_Icc hp)]
  constructor
  · intro h
    have hsum : π / p + π / q > π / 2 := by linarith
    have hmul : π * (1 / (p : ℝ) + 1 / q) > π * (1 / 2) := by
      convert hsum using 1 <=> field_simp
    exact lt_of_mul_lt_mul_left hmul pi_pos.le
  · intro h
    have hmul : π * (1 / (p : ℝ) + 1 / q) > π * (1 / 2) :=
      mul_lt_mul_of_pos_left h pi_pos
    have hsum : π / p + π / q > π / 2 := by
      convert hmul using 1 <=> field_simp
    linarith

/-- Leading 3×3 minor `D₃ > 0` iff `{p, q}` is Platonic. -/
theorem leadingMinor_three_pos_iff {p q r : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) :
  0 < leadingMinor 3 (by decide) p q r ↔ IsPlatonicPair p q := by
  rw [leadingMinor_three, platonic_pair_iff_reciprocal hp hq, sub_pos,
    sq_lt_sq (cos_angle_pos hq).le (sin_angle_pos hp).le]
  exact sin_gt_cos_iff_reciprocal hp hq

/-- Trailing 3×3 minor `> 0` iff the vertex figure `{q, r}` is Platonic. -/
theorem trailing3_pos_iff {p q r : ℕ} (hq : 3 ≤ q) (hr : 3 ≤ r) :
  0 < (trailing3 p q r).det ↔ IsPlatonicPair q r := by
  rw [trailing3_det, ← leadingMinor_three (p := q) (q := r) (r := p)]
  exact leadingMinor_three_pos_iff hq hr

```

A.14 PlatonicSolids/QuadForm.lean

Quadratic form of the Gram matrix (26 lines).

```

/-
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-/
import PlatonicSolids.GramMatrix

/-!
Paper §3.3. The quadratic form of the Schläfli Gram matrix.
-/

open scoped Matrix
open Real Matrix

private lemma sum_fin4 (f : Fin 4 → ℝ) :  $\sum i, f i = f 0 + f 1 + f 2 + f 3$  := by
  rw [Fin.sum_univ_succ, Fin.sum_univ_succ, Fin.sum_univ_succ]
  simp
  abel

lemma schlaefliGram_quad (p q r : ℕ) (x : Fin 4 → ℝ) :
  star x ·v (schlaefliGram p q r *v x) =
    x 0 ^ 2 + x 1 ^ 2 + x 2 ^ 2 + x 3 ^ 2
    - 2 * cos (π / p) * x 0 * x 1
    - 2 * cos (π / q) * x 1 * x 2
    - 2 * cos (π / r) * x 2 * x 3 := by
  simp [schlaefliGram, dotProduct, mulVec, sum_fin4]
  ring

```

A.15 PlatonicSolids/CompletingSquare.lean

LDL / completing-the-square identity (55 lines).

```

/-
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-/
import Mathlib.Tactic

/-!
Paper §3.3. Completing the square (LDL) for the rank-4 tridiagonal
quadratic form that the Schläfli Gram matrix induces.
-/

/-- Weighted sum of squares after the triangular change of variables. -/
lemma band_quad_squares (α β γ : ℝ) (x0 x1 x2 x3 : ℝ)
  (h2 : 1 - α ^ 2 ≠ 0) (h3 : 1 - α ^ 2 - β ^ 2 ≠ 0) :
  x0 ^ 2 + x1 ^ 2 + x2 ^ 2 + x3 ^ 2 + 2 * α * x0 * x1 + 2 * β * x1 * x2 +
    2 * γ * x2 * x3 =
    (x0 + α * x1) ^ 2
    + (1 - α ^ 2) * (x1 + β / (1 - α ^ 2) * x2) ^ 2
    + ((1 - α ^ 2 - β ^ 2) / (1 - α ^ 2)) *
      (x2 + γ * (1 - α ^ 2) / (1 - α ^ 2 - β ^ 2) * x3) ^ 2
    + ((1 - α ^ 2 - β ^ 2 - γ ^ 2 * (1 - α ^ 2)) / (1 - α ^ 2 - β ^ 2)) *
      x3 ^ 2 := by
  field_simp [h2, h3]
  ring

lemma weighted_squares_pos {a b c d c1 c2 c3 : ℝ}
  (hc1 : 0 < c1) (hc2 : 0 < c2) (hc3 : 0 < c3)
  (h : a ≠ 0 ∨ b ≠ 0 ∨ c ≠ 0 ∨ d ≠ 0) :
  0 < a ^ 2 + c1 * b ^ 2 + c2 * c ^ 2 + c3 * d ^ 2 := by
  rcases h with ha | hb | hc | hd
  · nlinarith [sq_pos_of_ne_zero ha, sq_nonneg b, sq_nonneg c, sq_nonneg d]
  · nlinarith [sq_pos_of_ne_zero hb, sq_nonneg a, sq_nonneg c, sq_nonneg d]

```

```

· nlinarith [sq_pos_of_ne_zero hc, sq_nonneg a, sq_nonneg b, sq_nonneg d]
· nlinarith [sq_pos_of_ne_zero hd, sq_nonneg a, sq_nonneg b, sq_nonneg c]

/-- The triangular change of variables is invertible. -/
lemma ldl_coords_eq_zero {α β γ x0 x1 x2 x3 : ℝ}
  (_h2 : 1 - α ^ 2 ≠ 0) (_h3 : 1 - α ^ 2 - β ^ 2 ≠ 0)
  (h01 : x0 + α * x1 = 0)
  (h12 : x1 + β / (1 - α ^ 2) * x2 = 0)
  (h23 : x2 + γ * (1 - α ^ 2) / (1 - α ^ 2 - β ^ 2) * x3 = 0)
  (h3z : x3 = 0) :
  x0 = 0 ∧ x1 = 0 ∧ x2 = 0 ∧ x3 = 0 := by
  have hx3 : x3 = 0 := h3z
  have hx2 : x2 = 0 := by
    rw [hx3] at h23
    simp using h23
  have hx1 : x1 = 0 := by
    rw [hx2] at h12
    simp at h12
    exact h12
  have hx0 : x0 = 0 := by
    rw [hx1] at h01
    simp using h01
  exact ⟨hx0, hx1, hx2, hx3⟩

```

A.16 PlatonicSolids/DetFormula.lean

$\det M = \Delta$ (43 lines).

```

/-
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-/
import Mathlib.LinearAlgebra.Matrix.Determinant.Basic
import PlatonicSolids.GramMatrix

/-!
Paper §3.4. The determinant of the rank-4 Schläfli Gram matrix is
 $\Delta = \sin^2(\pi/p) \sin^2(\pi/r) - \cos^2(\pi/q)$ .
-/

open scoped Matrix
open Real Matrix

private lemma sum_fin4 (f : Fin 4 → ℝ) :  $\sum i, f i = f 0 + f 1 + f 2 + f 3$  := by
  rw [Fin.sum_univ_succ, Fin.sum_univ_succ, Fin.sum_univ_succ]
  simp
  abel

/-- Determinant of the tridiagonal Gram pattern. -/
lemma det_band (a b c : ℝ) :
  (!! [1, a, 0, 0;
    a, 1, b, 0;
    0, b, 1, c;
    0, 0, c, 1] : Matrix (Fin 4) (Fin 4) ℝ).det =
  (1 - a ^ 2) * (1 - c ^ 2) - b ^ 2 := by
  rw [det_succ_row_zero, sum_fin4]
  simp [det_fin_three, Fin.succAbove]
  ring

lemma schlafliGram_det (p q r : ℕ) :
  (schlafliGram p q r).det = schlafliDet p q r := by
  unfold schlafliGram schlafliDet
  rw [det_band]

```

```

have hp : 1 - (-cos (π / p)) ^ 2 = sin (π / p) ^ 2 := by linarith [sin_sq_add_cos_sq (π / p)]
have hr : 1 - (-cos (π / r)) ^ 2 = sin (π / r) ^ 2 := by linarith [sin_sq_add_cos_sq (π / r)]
rw [hp, hr]
ring

lemma leadingMinor_four (p q r : ℕ) :
  (schlafliGram p q r).det = schlafliDet p q r :=
  schlafliGram_det p q r

```

A.17 PlatonicSolids/Sylvester.lean

Sylvester: positive definite iff leading minors are positive (100 lines).

```

/-
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-/
import Mathlib.Analysis.Matrix.PosDef
import PlatonicSolids.CompletingSquare
import PlatonicSolids.DetFormula
import PlatonicSolids.LearningMinors
import PlatonicSolids.QuadForm

/-!
Paper §3.3. Sylvester's criterion for the Schläfli Gram matrix:
positive-definite iff the four leading principal minors are positive.
`PosDef →` minors uses Mathlib (`PosDef.submatrix`, `PosDef.det_pos`);
the converse is the completing-the-square identity.
-/

open scoped Matrix
open Real Matrix

lemma leadingMinors_pos_of_posDef {p q r : ℕ} (h : (schlafliGram p q r).PosDef) :
  0 < leadingMinor 1 (by decide) p q r ∧
  0 < leadingMinor 2 (by decide) p q r ∧
  0 < leadingMinor 3 (by decide) p q r ∧
  0 < (schlafliGram p q r).det := by
  have h1 := h.submatrix (Fin.castLE_injective (by decide : 1 ≤ 4))
  have h2 := h.submatrix (Fin.castLE_injective (by decide : 2 ≤ 4))
  have h3 := h.submatrix (Fin.castLE_injective (by decide : 3 ≤ 4))
  refine ⟨?, ?, ?, PosDef.det_pos h⟩
  · simp [leadingMinor, leadingSubmatrix] using PosDef.det_pos h1
  · simp [leadingMinor, leadingSubmatrix] using PosDef.det_pos h2
  · simp [leadingMinor, leadingSubmatrix] using PosDef.det_pos h3

/-- Converse Sylvester for this Gram family: positive leading minors ⇒ `PosDef`. -/
theorem schlafliGram_posDef_of_minors {p q r : ℕ}
  (h1 : 0 < leadingMinor 1 (by decide) p q r)
  (h2 : 0 < leadingMinor 2 (by decide) p q r)
  (h3 : 0 < leadingMinor 3 (by decide) p q r)
  (h4 : 0 < (schlafliGram p q r).det) :
  (schlafliGram p q r).PosDef := by
  have hD1 : leadingMinor 1 (by decide) p q r = 1 := leadingMinor_one p q r
  have : (0 : ℝ) < 1 := hD1 ▸ h1
  clear this
  refine posDef_iff_dotProduct_mulVec.mpr ⟨schlafliGram_isHermitian p q r, ?_⟩
  intro x hx
  set α := -cos (π / p)
  set β := -cos (π / q)
  set γ := -cos (π / r)
  have hα : 1 - α ^ 2 = sin (π / p) ^ 2 := by
    simp [α]; linarith [sin_sq_add_cos_sq (π / p)]

```

```

have hβ : 1 - α ^ 2 - β ^ 2 = sin (π / p) ^ 2 - cos (π / q) ^ 2 := by
  simp [α, β]; linarith [sin_sq_add_cos_sq (π / p)]
have hD2 : leadingMinor 2 (by decide) p q r = sin (π / p) ^ 2 := leadingMinor_two p q r
have hD3 : leadingMinor 3 (by decide) p q r =
  sin (π / p) ^ 2 - cos (π / q) ^ 2 := leadingMinor_three p q r
have h2ne : 1 - α ^ 2 ≠ 0 := by
  rw [hα]; exact ne_of_gt (hD2 ► h2)
have h3ne : 1 - α ^ 2 - β ^ 2 ≠ 0 := by
  rw [hβ]; exact ne_of_gt (hD3 ► h3)
have hc1 : 0 < 1 - α ^ 2 := by rw [hα]; exact hD2 ► h2
have hc2 : 0 < 1 - α ^ 2 - β ^ 2 := by rw [hβ]; exact hD3 ► h3
have hγ : 1 - γ ^ 2 = sin (π / r) ^ 2 := by
  simp [γ]; linarith [sin_sq_add_cos_sq (π / r)]
have hc3 : 0 < 1 - α ^ 2 - β ^ 2 - γ ^ 2 * (1 - α ^ 2) := by
  have : 1 - α ^ 2 - β ^ 2 - γ ^ 2 * (1 - α ^ 2) =
    (1 - α ^ 2) * (1 - γ ^ 2) - β ^ 2 := by ring
  rw [this, hα, hγ]
have hβsq : β ^ 2 = cos (π / q) ^ 2 := by simp [β]
rw [hβsq]
have hdet : sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2 =
  (schlaflGram p q r).det := by
  rw [schlaflGram_det, schlaflDet]
rw [hdet]
exact h4
have hQ : star x ·v (schlaflGram p q r ·v x) =
  (x 0 + α * x 1) ^ 2
  + (1 - α ^ 2) * (x 1 + β / (1 - α ^ 2) * x 2) ^ 2
  + ((1 - α ^ 2 - β ^ 2) / (1 - α ^ 2)) *
    (x 2 + γ * (1 - α ^ 2) / (1 - α ^ 2 - β ^ 2) * x 3) ^ 2
  + ((1 - α ^ 2 - β ^ 2 - γ ^ 2 * (1 - α ^ 2)) / (1 - α ^ 2 - β ^ 2)) *
    x 3 ^ 2 := by
  calc star x ·v (schlaflGram p q r ·v x)
    = x 0 ^ 2 + x 1 ^ 2 + x 2 ^ 2 + x 3 ^ 2
      - 2 * cos (π / p) * x 0 * x 1
      - 2 * cos (π / q) * x 1 * x 2
      - 2 * cos (π / r) * x 2 * x 3 := schlaflGram_quad p q r x
    _ = x 0 ^ 2 + x 1 ^ 2 + x 2 ^ 2 + x 3 ^ 2
      + 2 * α * x 0 * x 1 + 2 * β * x 1 * x 2 + 2 * γ * x 2 * x 3 := by
      simp [α, β, γ]; ring
    _ = _ := band_quad_squares α β γ (x 0) (x 1) (x 2) (x 3) h2ne h3ne
rw [hQ]
refine weighted_squares_pos hc1 (div_pos hc2 hc1) (div_pos hc3 hc2) ?_
by_contra hforms
push Not at hforms
obtain ⟨h01, h12, h23, h3z⟩ := hforms
obtain ⟨hx0, hx1, hx2, hx3⟩ := ldl_coords_eq_zero h2ne h3ne h01 h12 h23 h3z
have : x = 0 := by
  ext i
  fin_cases i <|> simp [hx0, hx1, hx2, hx3]
exact hx this

```

A.18 PlatonicSolids/TrigValues.lean

Exact
 $\sin^2(\frac{\pi}{n})$ and
 $\cos^2(\frac{\pi}{n})$ for n
in
3, 4, 5 (63 lines).

```

/-
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-/
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic

/-!
Paper §4. Exact  $\sin^2(\pi/n)$  and  $\cos^2(\pi/n)$  for  $n \in \{3,4,5\}$ , from
Mathlib's sin_pi_div_* / cos_pi_div_*.
-/

open Real

lemma sq_cos_pi_div_three :  $\cos(\pi / 3)^2 = 1 / 4$  := by
  rw [cos_pi_div_three]; norm_num

lemma sq_sin_pi_div_four :  $\sin(\pi / 4)^2 = 1 / 2$  := by
  rw [sin_pi_div_four, div_pow, sq_sqrt (by norm_num : (0 : ℝ) ≤ 2)]; norm_num

lemma sq_cos_pi_div_four :  $\cos(\pi / 4)^2 = 1 / 2$  := by
  rw [cos_pi_div_four, div_pow, sq_sqrt (by norm_num : (0 : ℝ) ≤ 2)]; norm_num

lemma sq_cos_pi_div_five :  $\cos(\pi / 5)^2 = (3 + \sqrt{5}) / 8$  := by
  rw [cos_pi_div_five]
  have :  $(1 + \sqrt{5})^2 = 6 + 2 * \sqrt{5}$  := by
    ring_nf
    rw [sq_sqrt (by norm_num : (0 : ℝ) ≤ 5)]
    ring
  field_simp
  rw [this]
  ring

lemma sq_sin_pi_div_five :  $\sin(\pi / 5)^2 = (5 - \sqrt{5}) / 8$  := by
  have h :  $\sin(\pi / 5)^2 = 1 - \cos(\pi / 5)^2$  := by
    linarith [sin_sq_add_cos_sq (pi / 5)]
  rw [h, sq_cos_pi_div_five]
  field_simp
  ring

lemma sqrt5_bounds :  $2 < \sqrt{5} \wedge \sqrt{5} < 3$  := by
  constructor
  · rw [lt_sqrt (by norm_num)]; norm_num
  · rw [sqrt_lt' (by norm_num)]; norm_num

lemma three_mul_sqrt5_lt_seven :  $3 * \sqrt{5} < 7$  := by
  have h :  $(3 * \sqrt{5})^2 < (7 : ℝ)^2$  := by
    calc (3 *  $\sqrt{5}$ )^2 = 9 * ( $\sqrt{5}$ ^2) := by ring
    _ = 9 * 5 := by rw [sq_sqrt (by norm_num)]
    _ = 45 := by norm_num
    _ < 49 := by norm_num
    _ = (7 : ℝ)^2 := by norm_num
  have :  $0 \leq 3 * \sqrt{5}$  := by positivity
  nlinarith

lemma two_mul_sqrt5_gt_three :  $3 < 2 * \sqrt{5}$  := by
  have h :  $(3 : ℝ)^2 < (2 * \sqrt{5})^2$  := by
    calc (3 : ℝ)^2 = 9 := by norm_num
    _ < 20 := by norm_num
    _ = 4 * 5 := by norm_num
    _ = 4 * ( $\sqrt{5}$ ^2) := by rw [sq_sqrt (by norm_num)]
    _ = (2 *  $\sqrt{5}$ )^2 := by ring

```

```

have : 0 ≤ 2 * √5 := by positivity
nlinarith

```

A.19 PlatonicSolids/CompatibleTriples.lean

The eleven Platonic-compatible Schläfli triples (64 lines).

```

/-
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-/
import PlatonicSolids.PlatonicPair

/-!
Paper §4. The eleven Schläfli triples whose cell and vertex figure
are both Platonic.
-/

/-- Compatible middle index: 11 of the 25 Platonic-pair products survive. -/
theorem platonic_compatible_eleven {p q r : ℕ}
  (hc : IsPlatonicPair p q) (hv : IsPlatonicPair q r) :
  p = 3 ∧ q = 3 ∧ r = 3 ∨
  p = 3 ∧ q = 3 ∧ r = 4 ∨
  p = 3 ∧ q = 3 ∧ r = 5 ∨
  p = 3 ∧ q = 4 ∧ r = 3 ∨
  p = 4 ∧ q = 3 ∧ r = 3 ∨
  p = 4 ∧ q = 3 ∧ r = 4 ∨
  p = 4 ∧ q = 3 ∧ r = 5 ∨
  p = 3 ∧ q = 5 ∧ r = 3 ∨
  p = 5 ∧ q = 3 ∧ r = 3 ∨
  p = 5 ∧ q = 3 ∧ r = 4 ∨
  p = 5 ∧ q = 3 ∧ r = 5 := by
  rcases hc with ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩ | ⟨rfl, rfl⟩
  · -- cell {3,3}
    rcases hv with ⟨_, rfl⟩ | ⟨_, rfl⟩ | ⟨h, _⟩ | ⟨_, rfl⟩ | ⟨h, _⟩
    · exact .inl ⟨rfl, rfl, rfl⟩
    · exact .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
    · exact .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
  · -- cell {3,4}
    rcases hv with ⟨h, _⟩ | ⟨h, _⟩ | ⟨_, rfl⟩ | ⟨h, _⟩ | ⟨h, _⟩
    · cases h
    · cases h
    · exact .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
    · cases h
  · -- cell {4,3}
    rcases hv with ⟨_, rfl⟩ | ⟨_, rfl⟩ | ⟨h, _⟩ | ⟨_, rfl⟩ | ⟨h, _⟩
    · exact .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
    · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
    · cases h
  · -- cell {3,5}
    rcases hv with ⟨h, _⟩ | ⟨h, _⟩ | ⟨h, _⟩ | ⟨h, _⟩ | ⟨_, rfl⟩
    · cases h
    · cases h
    · cases h
    · cases h
    · exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
  · -- cell {5,3}
    rcases hv with ⟨_, rfl⟩ | ⟨_, rfl⟩ | ⟨h, _⟩ | ⟨_, rfl⟩ | ⟨h, _⟩

```

```

· exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <|
  .inl ⟨rfl, rfl, rfl⟩
· exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <|
  .inr <| .inl ⟨rfl, rfl, rfl⟩
· cases h
· exact .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <| .inr <|
  .inr <| .inl ⟨rfl, rfl, rfl⟩
· cases h

```

A.20 PlatonicSolids/SignTable.lean

Sign of Δ on those eleven triples (121 lines).

```

/-
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-/
import PlatonicSolids.GramMatrix
import PlatonicSolids.TrigValues

/-!
Paper §4. Sign of `Δ` on the eleven Platonic-compatible triples.
-/

open Real

private lemma unfold_det (p q r : ℕ) :
  schlaflDet p q r = sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2 :=
  rfl

private lemma cast_angles :
  sin (π / (3 : ℕ)) ^ 2 = sin (π / 3) ^ 2 ∧
  sin (π / (4 : ℕ)) ^ 2 = sin (π / 4) ^ 2 ∧
  sin (π / (5 : ℕ)) ^ 2 = sin (π / 5) ^ 2 ∧
  cos (π / (3 : ℕ)) ^ 2 = cos (π / 3) ^ 2 ∧
  cos (π / (4 : ℕ)) ^ 2 = cos (π / 4) ^ 2 ∧
  cos (π / (5 : ℕ)) ^ 2 = cos (π / 5) ^ 2 := by
  simp [Nat.cast_ofNat]

lemma schlaflDet_333 : schlaflDet 3 3 3 = 5 / 16 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.2.2.1, sq_sin_pi_div_three, sq_cos_pi_div_three]
  norm_num

lemma schlaflDet_334 : schlaflDet 3 3 4 = 1 / 8 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.1, cast_angles.2.2.2.1,
    sq_sin_pi_div_three, sq_sin_pi_div_four, sq_cos_pi_div_three]
  norm_num

lemma schlaflDet_433 : schlaflDet 4 3 3 = 1 / 8 := by
  rw [unfold_det, cast_angles.2.1, cast_angles.1, cast_angles.2.2.2.1,
    sq_sin_pi_div_four, sq_sin_pi_div_three, sq_cos_pi_div_three]
  norm_num

lemma schlaflDet_343 : schlaflDet 3 4 3 = 1 / 16 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.2.2.2.1, sq_sin_pi_div_three, sq_cos_pi_div_four]
  ]
  norm_num

lemma schlaflDet_335 : schlaflDet 3 3 5 = (7 - 3 * √5) / 32 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.2.1, cast_angles.2.2.2.1,
    sq_sin_pi_div_three, sq_sin_pi_div_five, sq_cos_pi_div_three]
  ring

```

```

lemma schlaflDet_533 : schlaflDet 5 3 3 = (7 - 3 * √5) / 32 := by
  rw [unfold_det, cast_angles.2.2.1, cast_angles.1, cast_angles.2.2.2.1,
      sq_sin_pi_div_five, sq_sin_pi_div_three, sq_cos_pi_div_three]
  ring

lemma schlaflDet_434 : schlaflDet 4 3 4 = 0 := by
  rw [unfold_det, cast_angles.2.1, cast_angles.2.2.2.1, sq_sin_pi_div_four, sq_cos_pi_div_three]
  ]
  norm_num

lemma schlaflDet_353 : schlaflDet 3 5 3 = (3 - 2 * √5) / 16 := by
  rw [unfold_det, cast_angles.1, cast_angles.2.2.2.2, sq_sin_pi_div_three, sq_cos_pi_div_five]
  ]
  ring

lemma schlaflDet_435 : schlaflDet 4 3 5 = (1 - √5) / 16 := by
  rw [unfold_det, cast_angles.2.1, cast_angles.2.2.1, cast_angles.2.2.2.1,
      sq_sin_pi_div_four, sq_sin_pi_div_five, sq_cos_pi_div_three]
  ring

lemma schlaflDet_534 : schlaflDet 5 3 4 = (1 - √5) / 16 := by
  rw [unfold_det, cast_angles.2.2.1, cast_angles.2.1, cast_angles.2.2.2.1,
      sq_sin_pi_div_five, sq_sin_pi_div_four, sq_cos_pi_div_three]
  ring

lemma schlaflDet_535 : schlaflDet 5 3 5 = (14 - 10 * √5) / 64 := by
  rw [unfold_det, cast_angles.2.2.1, cast_angles.2.2.2.1, sq_sin_pi_div_five,
      sq_cos_pi_div_three]
  have : ((5 - √5) / 8) ^ 2 = (30 - 10 * √5) / 64 := by
    field_simp
    ring_nf
    rw [sq_sqrt (by norm_num : (0 : ℝ) ≤ 5)]
    ring
  rw [← pow_two, this]
  ring

lemma schlaflDet_333_pos : 0 < schlaflDet 3 3 3 := by rw [schlaflDet_333]; norm_num
lemma schlaflDet_334_pos : 0 < schlaflDet 3 3 4 := by rw [schlaflDet_334]; norm_num
lemma schlaflDet_433_pos : 0 < schlaflDet 4 3 3 := by rw [schlaflDet_433]; norm_num
lemma schlaflDet_343_pos : 0 < schlaflDet 3 4 3 := by rw [schlaflDet_343]; norm_num

lemma schlaflDet_335_pos : 0 < schlaflDet 3 3 5 := by
  rw [schlaflDet_335]
  have : 0 < 7 - 3 * √5 := by linarith [three_mul_sqrt5_lt_seven]
  positivity

lemma schlaflDet_533_pos : 0 < schlaflDet 5 3 3 := by
  rw [schlaflDet_533]
  have : 0 < 7 - 3 * √5 := by linarith [three_mul_sqrt5_lt_seven]
  positivity

lemma schlaflDet_434_nonpos : schlaflDet 4 3 4 ≤ 0 := by
  rw [schlaflDet_434]

lemma schlaflDet_353_neg : schlaflDet 3 5 3 < 0 := by
  rw [schlaflDet_353]
  have : 3 - 2 * √5 < 0 := by linarith [two_mul_sqrt5_gt_three]
  exact div_neg_of_neg_of_pos this (by positivity)

lemma schlaflDet_435_neg : schlaflDet 4 3 5 < 0 := by
  rw [schlaflDet_435]
  have : 1 - √5 < 0 := by linarith [sqrt5_bounds.1]

```

```

exact div_neg_of_neg_of_pos this (by positivity)

lemma schlaflidDet_534_neg : schlaflidDet 5 3 4 < 0 := by
  rw [schlaflidDet_534]
  have : 1 -  $\sqrt{5}$  < 0 := by linarith [sqrt5_bounds.1]
  exact div_neg_of_neg_of_pos this (by positivity)

lemma schlaflidDet_535_neg : schlaflidDet 5 3 5 < 0 := by
  rw [schlaflidDet_535]
  have : 14 - 10 *  $\sqrt{5}$  < 0 := by
    have : 14 < 10 *  $\sqrt{5}$  := by nlinarith [sqrt5_bounds.1]
    linarith
  exact div_neg_of_neg_of_pos this (by positivity)

```

A.21 PlatonicSolids/Regular4.lean

The six regular convex 4-polytopes (28 lines).

```

/-
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-/
import PlatonicSolids.PlatonicPair

/-!
Paper §4, Group I. The six regular convex 4-polytopes.
-/

/-- Schläfli triples of the six regular convex 4-polytopes. -/
def IsRegular4Polytope (p q r : ℕ) : Prop :=
  (p = 3 ∧ q = 3 ∧ r = 3) ∨
  (p = 4 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 4) ∨
  (p = 3 ∧ q = 4 ∧ r = 3) ∨
  (p = 5 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 5)

theorem IsRegular4Polytope.ge_three {p q r : ℕ} (h : IsRegular4Polytope p q r) :
  3 ≤ p ∧ 3 ≤ q ∧ 3 ≤ r := by
  rcases h with ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ <;> decide

theorem IsRegular4Polytope.platonic {p q r : ℕ} (h : IsRegular4Polytope p q r) :
  IsPlatonicPair p q ∧ IsPlatonicPair q r := by
  rcases h with ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ <;> simp [IsPlatonicPair]

```

A.22 PlatonicSolids/Classification.lean

$\Delta > 0$ (equivalently M positive definite) iff one of the six (84 lines).

```

/-
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-/
import PlatonicSolids.Angles
import PlatonicSolids.CompatibleTriples
import PlatonicSolids.D3IffPlatonic
import PlatonicSolids.Regular4
import PlatonicSolids.SignTable
import PlatonicSolids.Sylvester
import PlatonicSolids.TrailingMinor

```

```

/-!
Paper §4. The six regular convex 4-polytopes are exactly the
Platonic-compatible triples with  $\Delta > 0$ , equivalently those whose
Schläfli Gram matrix is positive definite.
-/

open Real Matrix

/-- Palomar compared 4D theorem: Platonic cell and vertex figure with
 $\Delta > 0$  force one of the six regular convex 4-polytopes. -/
theorem regular_polychora_classification (p q r : ℕ)
  (h_cell : IsPlatonicPair p q)
  (h_vf : IsPlatonicPair q r)
  (h_det : 0 < schlaflDet p q r) :
  IsRegular4Polytope p q r := by
  rcases platonic_compatible_eleven h_cell h_vf with
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  · exact .inl ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inr <| .inr <| .inr <| .inr ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
  · exact .inr <| .inl ⟨rfl, rfl, rfl⟩
  · exact (not_lt_of_ge schlaflDet_434_nonpos h_det).elim
  · exact (lt_asymm schlaflDet_435_neg h_det).elim
  · exact (lt_asymm schlaflDet_353_neg h_det).elim
  · exact .inr <| .inr <| .inr <| .inr <| .inl ⟨rfl, rfl, rfl⟩
  · exact (lt_asymm schlaflDet_534_neg h_det).elim
  · exact (lt_asymm schlaflDet_535_neg h_det).elim

theorem schlaflDet_pos_of_regular {p q r : ℕ} (h : IsRegular4Polytope p q r) :
  0 < schlaflDet p q r := by
  rcases h with ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩ | ⟨rfl, rfl, rfl⟩
  · exact schlaflDet_333_pos
  · exact schlaflDet_433_pos
  · exact schlaflDet_334_pos
  · exact schlaflDet_343_pos
  · exact schlaflDet_533_pos
  · exact schlaflDet_335_pos

theorem regular_of_posDef {p q r : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) (hr : 3 ≤ r)
  (h : (schlaflGram p q r).PosDef) :
  IsRegular4Polytope p q r := by
  obtain ⟨_, _, h3, h4⟩ := leadingMinors_pos_of_posDef h
  have hcell : IsPlatonicPair p q := (leadingMinor_three_pos_iff hp hq).mp h3
  have hvf : IsPlatonicPair q r := by
    have hs := h.submatrix (Fin.succ_injective (n := 3))
    have : 0 < (trailing3 p q r).det := by
      simpa [trailing3] using PosDef.det_pos hs
    exact (trailing3_pos_iff hq hr).mp this
  have hdet : 0 < schlaflDet p q r := by
    rw [← schlaflGram_det]
    exact PosDef.det_pos h
  exact regular_polychora_classification p q r hcell hvf hdet

theorem posDef_of_regular {p q r : ℕ} (h : IsRegular4Polytope p q r) :
  (schlaflGram p q r).PosDef := by
  obtain ⟨hp, hq, _hr⟩ := h.ge_three
  obtain ⟨hcell, _hvf⟩ := h.platonic
  refine schlaflGram_posDef_of_minors ?_ ?_ ?_

```

```

· rw [leadingMinor_one]; norm_num
· rw [leadingMinor_two]; exact sq_pos_of_pos (sin_angle_pos hp)
· exact (leadingMinor_three_pos_iff hp hq).mpr hcell
· rw [schlafliGram_det]; exact schlafliDet_pos_of_regular h

/-- Finite regular convex 4-polytopes are exactly the triples `{p,q,r}`
with `{p,q,r} ≥ 3` whose Schläfli Gram matrix is positive definite. -/
theorem regular_polychora_iff {p q r : ℕ} (hp : 3 ≤ p) (hq : 3 ≤ q) (hr : 3 ≤ r) :
  (schlafliGram p q r).PosDef ↔ IsRegular4Polytope p q r :=
  ⟨regular_of_posDef hp hq hr, posDef_of_regular⟩

```

A.23 Challenge.lean

Palomar statement of record (deliberate sorries) (113 lines).

```

/-
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-/
import Mathlib.Analysis.SpecialFunctions.Trigonometric.Basic
import Mathlib.LinearAlgebra.Matrix.Notation
import Mathlib.Tactic

/-!
# Palomar statement of record (Platonic solids and regular polychora)

This module states the compared Palomar family and the Mathlib type
surface it depends on. Challenge may use `sorry`; Solution supplies the
proofs.

The selected theorems are:

1. `platonic_solids_3d` - Euler  $(V-E+F=2)$  plus regularity incidence
 $(pF=2E)$ ,  $(qV=2E)$  force a Platonic pair  $((p,q))$ .
2. `edges_pos_of_regular` - the same identities algebraically force
 $(E>0)$ ; non-degeneracy is a theorem, not an axiom.
3. `regular_polychora_classification` - Platonic cells and vertex figures
with positive Schläfli Gram determinant are exactly the six regular
convex 4-polytopes.

## How to read this file

The definitions below are the vocabulary of the claim. A reader who wants
to check what has been proved should read this file and need not read
anything else. `Solution.lean` imports the sorry-free development in
`PlatonicSolids.lean`. Each supporting lemma lives in a section-sized
module under `PlatonicSolids/`, meant to be read next to the matching
paragraph of `PlatonicSolids.md`.
-/

open scoped Matrix
open Real Matrix

/-- The 5 Platonic pairs (p, q):
(3, 3): Tetrahedron
(3, 4): Octahedron
(4, 3): Cube
(3, 5): Icosahedron
(5, 3): Dodecahedron -/
def IsPlatonicPair (p q : ℕ) : Prop :=
  (p = 3 ∧ q = 3) ∨
  (p = 3 ∧ q = 4) ∨

```

```

(p = 4 ∧ q = 3) ∨
(p = 3 ∧ q = 5) ∨
(p = 5 ∧ q = 3)

/-- Canonical bilinear form of the rank-4 Schläfli Coxeter system `{p, q, r}`. -/
noncomputable def schlaflGram (p q r : ℕ) : Matrix (Fin 4) (Fin 4) ℝ :=
  !![1, -cos (π / p), 0, 0;
    -cos (π / p), 1, -cos (π / q), 0;
    0, -cos (π / q), 1, -cos (π / r);
    0, 0, -cos (π / r), 1]

/-- `Δ(p, q, r) = sin²(π/p) sin²(π/r) - cos²(π/q)`. -/
noncomputable def schlaflDet (p q r : ℕ) : ℝ :=
  sin (π / p) ^ 2 * sin (π / r) ^ 2 - cos (π / q) ^ 2

/-- The 6 regular convex 4-polytopes (Schläfli symbols {p, q, r}):
  {3, 3, 3}: 5-cell (4D Simplex)
  {4, 3, 3}: 8-cell (Tesseract / Hypercube)
  {3, 3, 4}: 16-cell (Cross-polytope)
  {3, 4, 3}: 24-cell (Exceptional 4D regular polytope)
  {5, 3, 3}: 120-cell
  {3, 3, 5}: 600-cell -/
def IsRegular4Polytope (p q r : ℕ) : Prop :=
  (p = 3 ∧ q = 3 ∧ r = 3) ∨
  (p = 4 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 4) ∨
  (p = 3 ∧ q = 4 ∧ r = 3) ∨
  (p = 5 ∧ q = 3 ∧ r = 3) ∨
  (p = 3 ∧ q = 3 ∧ r = 5)

/-- Theorem: A Platonic configuration cannot have zero edges.
  The topological Euler characteristic ( $\chi = 2$ ) and regularity conditions ( $p, q \geq 3$ )
  algebraically force  $E \geq 1$  without requiring any geometric non-degeneracy axiom. -/
theorem edges_pos_of_regular
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  0 < E := by
  sorry

/-- The 3D Main Theorem:
  Any polyhedron satisfying Euler's formula ( $V - E + F = 2$ )
  and regularity incidence relations ( $pF = 2E, qV = 2E$ )
  must have (p, q) belonging to the 5 Platonic solids. -/
theorem platonic_solids_3d
  (V E F p q : ℕ)
  (hp : 3 ≤ p) (hq : 3 ≤ q)
  (h_edges_faces : p * F = 2 * E)
  (h_edges_verts : q * V = 2 * E)
  (h_euler : V + F = E + 2) :
  IsPlatonicPair p q := by
  sorry

/-- The Main Classification Theorem for 4D Regular Polychora:
  Any triple {p, q, r} whose 3D cells {p, q} and vertex figures {q, r} are
  Platonic solids, and whose Coxeter Gram matrix is positive-definite ( $\Delta > 0$ ),
  is strictly one of the 6 regular convex 4-polytopes. -/
theorem regular_polychora_classification (p q r : ℕ)
  (h_cell : IsPlatonicPair p q)
  (h_vf : IsPlatonicPair q r)
  (h_det : 0 < schlaflDet p q r) :
```

```

    IsRegular4Polytope p q r := by
sorry

```

A.24 Solution.lean

Palomar solution module: imports PlatonicSolids (20 lines).

```

/-
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Authors: Lars Warren Ericson.
-/
import PlatonicSolids

/-!
# Solutions to the Challenge

The declarations of `Challenge.lean`, proved. Importing `PlatonicSolids`
supplies `platonic_solids_3d`, `edges_pos_of_regular`,
`regular_polychora_classification` and the definitions they depend on
(`IsPlatonicPair`, `IsRegular4Polytope`, `schlafliGram`, `schlafliDet`),
with the same names and types as in the Challenge module.
Comparator compares those declarations.

The proofs are split across section-sized modules under `PlatonicSolids/`.
There is no `sorry` in those files.
-/

```